

|                                             |                                            |
|---------------------------------------------|--------------------------------------------|
| <b>Klastoets 3</b>                          | <b>Class Test 3</b>                        |
| Kopiereg voorbehou                          | Copyright reserved                         |
| <b>Elektrisiteit en Elektronika EBN 122</b> | <b>Electricity and Electronics EBN 122</b> |
| 24 Oktober 2018                             | 24 October 2018                            |

| Toetsinligting - Test Information                                                             |                       |                                    |            |
|-----------------------------------------------------------------------------------------------|-----------------------|------------------------------------|------------|
| Maksimum punte:<br>Maximum marks:                                                             | 15                    | Volpunte:<br>Full marks:           | 16         |
| Duur van vraestel:<br>Duration of paper:                                                      | 30 min + (10 for OCR) | Oop-/Toeboek:<br>Open/Closed book: | Toe/Closed |
| Totale aantal bladsye (hierdie blad ingesluit):<br>Total number of pages (this page included) | 6                     |                                    |            |

| Important                                                                          | Belangrik                                                                     |
|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| 1. The examination regulations of the University of Pretoria apply.                | 1. Die eksamenregulasies van die Universiteit van Pretoria geld.              |
| 2. All questions must be answered on the Optical Character Recognition (OCR) form. | 2. Alle vrae moet op die Optiese Karakter Herkenning (OKH) vorm ingevul word. |
| 3. Read the instructions on the OCR form carefully.                                | 3. Neem noukeurig kennis van die instruksies op die OKH vorm.                 |
| 4. Where applicable, answers must include units.                                   | 4. Indien van toepassing, moet antwoorde eenhede insluit.                     |
| 5. Final answers must be written as real numbers and not as fractions.             | 5. Finale antwoorde moet as reële getalle geskryf word, en nie as breuke nie. |
| 6. Use 5 significant numbers to write irrational numbers.                          | 6. Gebruik 5 beduidende syfers om irrasionele getalle voor te stel.           |

|                    |                                          |
|--------------------|------------------------------------------|
| Lecturers/Dosente: | D. le Roux, J.P. de Villiers, J.P Jacobs |
|--------------------|------------------------------------------|

### INSTRUCTIONS

**Do step 0 in the first 5 minutes of the session. (-5 to 0 min)**

**Do step 1 in the first 30 minutes of the test. (0-30 min)**

**Do step 2 within the following 10 minutes. (30-40 min)**

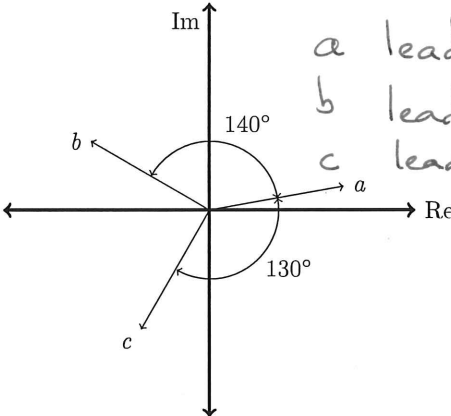
**STEP 0 (-5 min to 0 min):** Complete the front page of the OCR form.

**STEP 1 (0 min to 30 min):** Do your own calculations on the QUESTION PAPER. The question paper will not be taken in at the end of the test. Write down the answers for each question on the PRACTICE ANSWER SHEET. Be sure to include the units. The PRACTICE ANSWER SHEET will not be taken in at the end of the test.

**STEP 2 (30 min to 40 min):** Carefully copy the answers from the PRACTICE ANSWER SHEET to the OCR form.

- The Assessment ID is: **2018EBN122C3**
- Use a mechanical pencil with 0.5mm HB lead to complete the form.
- Each cell should only have one character, and characters may not touch or cross the cell.
- **Do not use commas!** Please make use of full stops as decimal points.

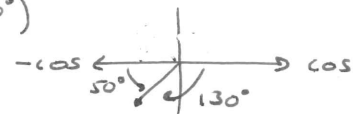
## Section 1 / Afdeling 1

| Question 1 [2]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Vraag 1 [2]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Given the diagram below, which statement is correct?</p> <ol style="list-style-type: none"> <li>Signal <math>c</math> lags <math>b</math> by <math>90^\circ</math> and leads <math>a</math> by <math>130^\circ</math>.</li> <li>Signal <math>c</math> leads <math>b</math> by <math>90^\circ</math> and leads <math>a</math> by <math>130^\circ</math>.</li> <li>Signal <math>b</math> leads <math>a</math> by <math>140^\circ</math> and lags <math>c</math> by <math>90^\circ</math>.</li> <li>Signal <math>b</math> lags <math>a</math> by <math>140^\circ</math> and lags <math>c</math> by <math>90^\circ</math>.</li> </ol> <p>(Write down the number of the correct statement as your answer.)</p> | <p>Gegewe onderstaande diagram, watter stelling is korrek?</p> <ol style="list-style-type: none"> <li>Sein <math>c</math> loop na <math>b</math> met <math>90^\circ</math> en voor <math>a</math> met <math>130^\circ</math>.</li> <li>Sein <math>c</math> loop voor <math>b</math> met <math>90^\circ</math> en voor <math>a</math> met <math>130^\circ</math>.</li> <li>Sein <math>b</math> loop voor <math>a</math> met <math>140^\circ</math> en na <math>c</math> met <math>90^\circ</math>.</li> <li>Sein <math>b</math> loop na <math>a</math> met <math>140^\circ</math> en na <math>c</math> met <math>90^\circ</math>.</li> </ol> <p>(Skryf die nommer neer van die korrekte stelling as jou antwoord.)</p> |
| 1 Statement #:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 1 Stelling #:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|  <p>Handwritten notes next to the diagram:</p> <ul style="list-style-type: none"> <li><math>a</math> leads <math>c</math> by <math>130^\circ</math></li> <li><math>b</math> leads <math>a</math> by <math>140^\circ</math></li> <li><math>c</math> leads <math>b</math> by <math>90^\circ</math></li> </ul>                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

| Question 2 [2]                                                                                                                                                                                                                                                              | Vraag 2 [2]                                                                                                                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>The following sinusoids are given:</p> $v_1(t) = -5 \cos(\omega t + 50^\circ) \text{ V}$ $v_2(t) = 5 \sin(\omega t + 50^\circ) \text{ V}$ <p>Determine the phasor value of <math>\bar{V}_T = \bar{V}_1 + \bar{V}_2</math>.<br/>Give the answer in <b>polar form</b>.</p> | <p>Die volgende sinusiede word gegee:</p> $v_1(t) = -5 \cos(\omega t + 50^\circ) \text{ V}$ $v_2(t) = 5 \sin(\omega t + 50^\circ) \text{ V}$ <p>Bepaal die fasor waarde van <math>\bar{V}_T = \bar{V}_1 + \bar{V}_2</math>.<br/>Gee die antwoord in <b>polêre vorm</b>.</p> |
| 2 $\bar{V}_T$                                                                                                                                                                                                                                                               | 2 $\bar{V}_T$                                                                                                                                                                                                                                                               |

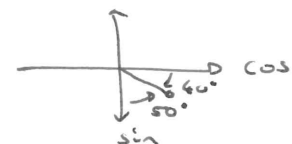
$$v_1 = -5 \cos(\omega t + 50^\circ) = 5 \cos(\omega t - 130^\circ)$$

$$\rightarrow \bar{V}_1 = 5 \angle -130^\circ$$



$$v_2 = 5 \sin(\omega t + 50^\circ) = 5 \cos(\omega t - 40^\circ)$$

$$\rightarrow \bar{V}_2 = 5 \angle -40^\circ$$

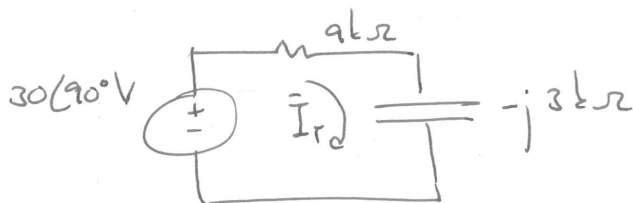


$$\therefore \bar{V}_T = 5 \angle -130^\circ + 5 \angle -40^\circ = 7.0711 \angle -85^\circ \text{ V}$$

| Question 3 [2]                                                                                                                                                                                                                                                                                                                   |             | Vraag 3 [2]                                                                                                                                                                                                                                                                                                                   |             |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| A series circuit consists of a $30 \cos(100 \times 10^3 t + 90^\circ)$ V voltage source, a $9 \text{ k}\Omega$ resistor, and a $3.333 \text{ nF}$ capacitor. Calculate the current $\bar{I}_T$ through the series circuit as it exists the positive terminal of the voltage source. Give the answer in <b>rectangular form</b> . |             | 'n Serie-kringbaan bestaan uit 'n $30 \cos(100 \times 10^3 t + 90^\circ)$ V spanningsbron, 'n $9 \text{ k}\Omega$ resistor, en 'n $3.333 \text{ nF}$ kapasitor. Bepaal die stroom $\bar{I}_T$ deur die serie kringbaan soos dit uit die positiewe terminaal van die bron vloeit. Gee die antwoord in <b>reghoekige vorm</b> . |             |
| 3                                                                                                                                                                                                                                                                                                                                | $\bar{I}_T$ | 3                                                                                                                                                                                                                                                                                                                             | $\bar{I}_T$ |

$$\omega = 100 \text{ krad/s.}$$

$$Z_c = \frac{-j}{100 \text{ k} \times 3.333 \text{ n}} = -j3 \text{ k}\Omega$$



$$\therefore \bar{I}_T = \frac{30 \angle 90^\circ}{(9 - j3) \text{ k}} = \underline{\underline{-1 + j3 \text{ mA}}}$$

| Question 4 [2]                                                                         |          | Vraag 4 [2]                                                                                 |          |
|----------------------------------------------------------------------------------------|----------|---------------------------------------------------------------------------------------------|----------|
| Find the equivalent impedance $Z_{ab}$ . Write the answer in <b>rectangular form</b> . |          | Bepaal die ekwivalente impedansie $Z_{ab}$ . Skryf die antwoord in <b>reghoekige vorm</b> . |          |
| 4                                                                                      | $Z_{ab}$ | 4                                                                                           | $Z_{ab}$ |
|                                                                                        |          |                                                                                             |          |

$$\begin{aligned}
 Z_{ab} &= -j3 \parallel (2 + j1) \\
 &= \frac{-j3(2 + j1)}{2 - j2} \\
 &= \frac{3 - j6}{2 - j2} \\
 &= \underline{\underline{2.25 - j0.75 \Omega}}
 \end{aligned}$$

| Question 5 [2]                                                  |             | Vraag 5 [2]                                                     |             |
|-----------------------------------------------------------------|-------------|-----------------------------------------------------------------|-------------|
| Determine $\bar{I}_x$ . Write the answer in <b>polar form</b> . |             | Bepaal $\bar{I}_x$ . Skryf die antwoord in <b>polêre vorm</b> . |             |
| 5                                                               | $\bar{I}_x$ | 5                                                               | $\bar{I}_x$ |

Current division:

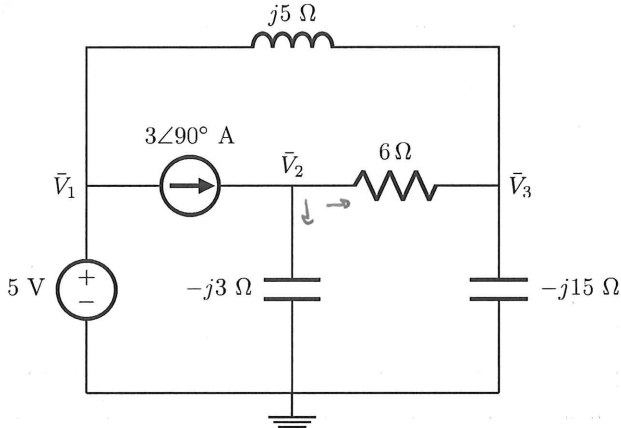
$$\begin{aligned}
 \bar{I}_x &= \frac{10\angle 50^\circ \times (5 - j15)}{5 - j15 + 10 + j60} = \frac{10\angle 50^\circ (5 - j15)}{15 + j45} \\
 &= \underline{3.333 \angle -93.13^\circ \text{ A}}
 \end{aligned}$$

| Question 6 [2]                                                     |                | Vraag 6 [2]                                                        |                |
|--------------------------------------------------------------------|----------------|--------------------------------------------------------------------|----------------|
| Determine $\bar{V}_{ab}$ . Write the answer in <b>polar form</b> . |                | Bepaal $\bar{V}_{ab}$ . Skryf die antwoord in <b>polêre vorm</b> . |                |
| 6                                                                  | $\bar{V}_{ab}$ | 6                                                                  | $\bar{V}_{ab}$ |

KVL:  $-j5 + 2\bar{I}_x + j3\bar{I}_x + 3\bar{I}_x - j8\bar{I}_x = 0.$

$$\therefore \bar{I}_x = \frac{j5}{5 - j5} = \frac{j}{1 - j} = -0.5 + j0.5 \text{ A.}$$

$$\begin{aligned}
 \therefore \bar{V}_{ab} &= j3\bar{I}_x + 3\bar{I}_x = (-0.5 + j0.5)(3 + j3) \\
 &= \underline{-3 \text{ V}}
 \end{aligned}$$

| Questions 7 to 10 [4]                                                                                                                                                                                                 |   |   |   | Vrae 7 tot 10 [4]                                                                                                                                                                                                             |   |    |   |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|---|
| Use nodal analysis to set up the following two equations of the form:<br>$a\bar{V}_2 + b\bar{V}_3 = -18$<br>$c\bar{V}_2 + d\bar{V}_3 = 30$<br><br>Calculate the unknown coefficients and write in rectangular format. |   |   |   | Gebruik nodelanalyse om die volgende twee vergelykings op te stel met die vorm:<br>$a\bar{V}_2 + b\bar{V}_3 = -18$<br>$c\bar{V}_2 + d\bar{V}_3 = 30$<br><br>Bepaal die onbekende koëffisiënte en skryf in reghoekige formaat. |   |    |   |
| 7                                                                                                                                                                                                                     | a | 8 | b | 9                                                                                                                                                                                                                             | c | 10 | d |
|                                                                                                                                     |   |   |   |                                                                                                                                                                                                                               |   |    |   |

$$* V_1 = 5V$$

$$\text{Node 2: } * 3\angle 90^\circ = \frac{V_2}{-j3} + \frac{V_2 - V_3}{6}$$

$$(\times j6) \quad j3 \times j6 = -2V_2 + jV_2 - jV_3$$

$$(-2 + j)V_2 - jV_3 = -18. \quad \longrightarrow$$

$$\text{Node 3: } * \frac{V_3}{-j15} + \frac{V_3 - V_2}{6} + \frac{V_3 - V_1}{j5} = 0$$

$$\times j30 \quad -2V_3 + j5V_3 - j5V_2 + 6V_3 - 6(5) = 0.$$

$$-j5V_2 + (4 + j5)V_3 = 30 \quad \longrightarrow$$

|                                   |
|-----------------------------------|
| Grand Total / Groot totaal : [16] |
|-----------------------------------|

**PRACTICE PAGE / OEFENBLAD**

|               |   |   |   |   |   |   |   |   |   |   |   |   |  |  |
|---------------|---|---|---|---|---|---|---|---|---|---|---|---|--|--|
| Assessment ID | 2 | 0 | 1 | 8 | E | B | N | 1 | 2 | 2 | C | 3 |  |  |
|---------------|---|---|---|---|---|---|---|---|---|---|---|---|--|--|

| How to enter complex numbers: |   |   |   |   |   |   |   |   |   |   |
|-------------------------------|---|---|---|---|---|---|---|---|---|---|
| 5                             | ∠ | - | 9 | 0 | ° |   |   |   | V |   |
| 6                             | . | 9 | ∠ | 1 | 5 | . | 9 | ° | μ | A |
| 1                             | 2 | - | j | 8 |   |   |   |   |   |   |
| j                             | 1 | 5 |   |   |   |   |   |   | Ω |   |
| -                             | 1 | . | 5 | + | j | 8 | . | 2 | m | V |

**Rectangular:**  $j$  is at the start of the imaginary number.  
 The  $j$  is in the CENTRE of the cell.  
 The  $j$  does NOT cross cell boundaries!

**Polar:** The degree sign ° is directly after the angle value.

Answer (please round all answers to 5 significant digits)

Unit

|    |                                       |                   |
|----|---------------------------------------|-------------------|
| 01 | Statement : 3                         | No unit required. |
| 02 | $\bar{V}_T = 7.0711 \angle -85^\circ$ | V                 |
| 03 | $\bar{I}_T = -1 + j3$                 | mA                |
| 04 | $Z_{ab} = 2.25 - j0.75$               | Ω                 |
| 05 | $\bar{I}_x = 3.333 \angle -93.13$     | A                 |
| 06 | $\bar{V}_{ab} = -3$                   | V                 |
| 07 | $a = -2 + j$                          | No unit required. |
| 08 | $b = -j$                              | No unit required. |
| 09 | $c = -j5$                             | No unit required. |
| 10 | $d = 4 + j5$                          | No unit required. |